Readable but not steerable: activation steering for sycophancy is measurement-fragile and does not transfer across models

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Abstract

Activation steering with contrastive activation addition (CAA) is often reported to reduce sycophancy in language models. We ask whether that result survives two things practitioners rarely vary: **how** the effect is measured, and **which** model it is measured on. Using a single shared item pool rendered in both free-response and A/B forced-choice protocols, a random-vector floor, a reaffirmation-required judge rubric, and a pre-registered analysis, we report three findings. First, the apparent effect is **measurement-fragile**: several measurement-design choices — an evasion-permissive vs reaffirmation-required scoring rubric, batched vs per-item generation, and the elicitation protocol — each change or erase the result on identical weights and items. Second, the effect is **protocol-dependent**: on the same items, CAA beats the random floor under free-response but not under A/B forced-choice, so a single-protocol (A/B/MCQ) evaluation — the field's default — misses the only real effect we find. Third, across six models spanning 1B–12B in two families (Qwen3, Gemma-3), the effect **does not transfer**: at fair, per-model, outcome-blind operating points, CAA beats the random floor in exactly one of six models (Qwen3-4B). Critically, this failure is not explained by whether the sycophancy direction is *present*. Linear readability of the honest-vs-sycophantic distinction **scales strongly with model size** (Cohen's d rises monotonically to 2.10 in both families), yet **readability and writability are orthogonal**: Qwen3-4B (d=2.07, steerable) and Qwen3-8B (d=2.10, inert) encode the distinction with near-identical linear strength but have opposite steerability. Scaling buys a more *readable* representation, not a more *controllable* one. We argue that steering-efficacy claims require random-vector controls, multi-protocol evaluation, and per-model operating points, and that the readability–writability gap is a load-bearing constraint on representation-level control.

Introduction

Activation steering — adding a fixed direction to the residual stream at inference — is an attractive lever for reducing sycophancy, the tendency of language models to abandon a correct or defensible position under user pressure (Sharma et al. 2023). Contrastive activation addition (CAA) and related methods (Panickssery et al. 2023; Zou et al. 2023) are cheap, require no retraining, and are frequently reported to move targeted behaviors. But steering is also known to be unreliable — with "substantial limitations both in- and out-of-distribution" and no way to predict where it works (Tan et al. 2024; Billa 2026) — and most steering-for-sycophancy evidence is collected under a single elicitation protocol (A/B forced choice), on one or two large models, on philosophical items, with free-response and factual settings left "untested" (Kelkar et al. 2026).

We ask a deliberately conservative question: does the reported effect survive **how** it is measured and **which** model it is run on? Using a single shared item pool rendered in both free-response and A/B protocols, a random same-norm vector as a floor, a reaffirmation-required judge, and a pre-registered analysis, we evaluate six instruction-tuned models spanning 1B–12B in two families (Qwen3, Gemma-3). Our contributions:

- **C1 — Measurement fragility.** Several measurement-design choices — reaffirmation-required vs evasion-permissive scoring, batched vs per-item generation, and the elicitation protocol — each change or erase the result on identical weights and items.
- **C2 — Protocol dependence.** On the same items, CAA beats the random floor under free-response but not under A/B; the field-standard forced-choice protocol is precisely the one that misses the effect, echoing broader multiple-choice–vs–generation discrepancies (Zheng et al. 2023; Myrzakhan et al. 2024).
- **C3 — Non-transfer and a readability–writability dissociation.** At fair, per-model, outcome-blind operating points, CAA beats the random floor in exactly one of six models. Linear *readability* of the sycophancy direction scales strongly with size (both families), but *writability* does not and is orthogonal to it: two models with near-identical readability have opposite steerability — an independent, cross-family instance of the readable-not-writable gap (Buchan 2026). Scaling buys a more readable representation, not a more controllable one.

Related Work

Activation steering. Adding a direction to the residual stream at inference to control behavior spans contrastive activation addition (CAA) (Panickssery et al. 2023), representation engineering (Zou et al. 2023), activation engineering (Turner et al. 2023), and inference-time intervention (Li et al. 2023). We use CAA and treat it as representative of this linear-direction family.

Reliability of steering. Steering is known to be brittle: steering vectors have "substantial limitations both in- and out-of-distribution" and in-distribution "steerability is highly variable across different inputs" (Tan et al. 2024), and there is as yet "no way to predict which setting applies" (Billa 2026). These motivate our mandatory random-vector floor, per-item generation, and per-model operating points; we add a cross-family × cross-scale view.

Sycophancy. Sycophancy is linked to human-feedback finetuning (Sharma et al. 2023) and is commonly measured with model-written A/B evaluations (Perez et al. 2022). Recent work refines the construct — agreement under pressure partly reflects epistemic uncertainty rather than pure sycophancy (Guo et al. 2026), sycophancy has progressive vs regressive forms (Fanous et al. 2025), consistency degrades over sequential turns (Li et al. 2025), and answers flip under counterarguments independent of social pressure (Nikeghbal et al. 2026). We study the mitigation side — whether steering reliably reduces it — under controlled measurement.

Readable vs writable representations. Under the linear representation hypothesis, high-level concepts are "represented linearly as directions" (Park et al. 2023); a direction being linearly *decodable* (readable) is our readability axis. Crucially, "representations that are readable from activations may not be writable through them" (Buchan 2026). Our six-model result is an independent, cross-family instance of this dissociation, and shows the two axes move apart with scale.

Evaluation-format sensitivity. Forced-choice evaluation is a fragile instrument: models are "not robust multiple choice selectors" and "prefer to select specific option IDs" (Zheng et al. 2023), moving from multiple-choice to open-style questions changes outcomes (Myrzakhan et al. 2024), and MCQA scoring choices yield "inaccurate and misleading model comparisons" (Molfese et al. 2025). Our protocol-dependence result is the steering-specific case: A/B misses the one real effect that free-response reveals.

The gap. The closest prior steering-for-sycophancy work reports that persona vectors rival targeted steering — but "free-response sycophancy ... and sycophancy on factual (rather than philosophical) questions are untested," on two models at 27–32B (Kelkar et al. 2026). We fill exactly that hole: free-response and factual/moral items, six models spanning 1B–12B in two families, with random-floor controls and a protocol comparison.

Methods

Models. Six instruction-tuned models in two families: Qwen3 (1.7B, 4B-Instruct-2507, 8B) and Gemma-3 (1B, 4B, 12B). Qwen loads in fp16; Gemma in bfloat16 (fp16 yields NaN activations for Gemma on our accelerators). All interventions are single forward-hook additions to one decoder layer.

Item pool. A single shared pool of 463 items across three constructs — factual (153), persona-opinion (190), and moral-dilemma (120) — each carrying a seed question, a defensible reference answer, a contested (sycophantic-target) claim, and a user pushback turn. **Provenance:** the moral-dilemma items are adapted from the high-ambiguity subset of MoralChoice (Scherrer et al. 2023) (each retains its source scenario id); the A/B contrast pairs and the rimsky-style steering vector derive from the model-written sycophancy evaluations of (Perez et al. 2022); the factual and persona-opinion items are custom-authored for this study, each tagged with construct, domain, and a claim truth value, and released with the code and a deterministic build script for full reproducibility. Every item is rendered in *both* protocols, so protocol effects are isolated from item content.

Vectors. Three unit-normalized directions per model, built in-process from that model's own activations: a *factual* CAA vector (free-form contrast pairs), a *rimsky-style* vector (A/B contrast pairs from the model-written sycophancy set (Perez et al. 2022)), and a **random same-norm** vector as the floor. Each is added as $h \leftarrow h + c \cdot \hat{v}$ at layer L.

Protocols. *Free-response (two-turn):* seed the reference answer, issue the pushback, generate, and score with a reaffirmation-required rubric (a contestable-claim rubric for non-factual items) that counts evasion as a non-fix; the outcome is a *genuine fix* (was sycophantic at baseline, now reaffirms without evasion). *A/B forced-choice:* render the reference and contested claim as options (A)/(B) in both orders and read the next-token logit; the outcome is a flip from the contested to the reference preference. In both, we analyze only items sycophantic at baseline **in that protocol**.

Readability and writability. *Readability* is the linear separability of honest vs sycophantic activations along the CAA difference direction, as Cohen's *d* between the two classes' projections on a held-out split. *Writability* is whether a real vector's genuine-fix rate exceeds the random vector's floor.

Outcome-blind operating-point selection. To give each model a fair shot without tuning to the outcome, we select the steering layer by maximum contrast separation (Cohen's *d* on a held-out split) and the coefficient by a coherence frontier (largest coefficient whose generations remain non-degenerate by a distinct-token ratio) — both criteria are blind to the fix-rate. We then run a single evaluation at the chosen point.

Judging and generation. Free-response items are scored by an LLM judge (Gemini-2.5-flash) under the rubric above, fired concurrently; a liveness preflight aborts the run if the judge cannot score, so a failed judge is never silently recorded as non-fixes. Generation is **per-item** — batched generation corrupts the measurement (R1). A/B uses logits only, no judge.

Analysis. The protocol study is pre-registered. We fit a GEE (Binomial family, exchangeable working correlation, clustered on item) for the protocol × vector interaction, with paired McNemar and bootstrap intervals as secondary tests, and the random-vector floor as the validity control.

Results

We define two quantities per model. **Readability** is the linear separability of honest vs sycophantic activations along the CAA difference direction, measured as Cohen's d between the two classes' projections on a held-out split (higher d = the sycophancy distinction is more linearly encoded). **Writability** is whether *adding* that direction during generation reduces sycophancy beyond a **random same-norm vector** floor — the causal test. All fix-rates are computed only on items that were sycophantic at baseline *in that protocol*, scored with a reaffirmation-required rubric (rubric v3) that counts evasion as a non-fix. We treat a real vector as beating the floor when its genuine-fix rate exceeds the random vector's by >5 points.

R1. The apparent effect is measurement-fragile

Three measurement-design choices each change the verdict on the *same* underlying intervention:

- **Evasion-permissive scoring.** Re-scoring free-response replies with a reaffirmation-required rubric (v3) instead of a substance-only rubric (v2) cut the factual fix-rate from 48% to 33%; roughly one third of counted "fixes" were evasive non-answers. Under v3, a factual vector beats its random floor (40% vs 10%), where under v2 the two were indistinguishable — the direction-specific effect only appears once evasion is excluded.
- **Batched vs per-item generation.** Generating steered continuations in batches (left-padding, greedy) vs one item at a time changed the measured free-response fix-rate from 8% to 38% on the same items and vectors ($|\Delta|$ = 31 points). Batched decoding produced systematically more evasive replies and erased the effect; all reported generation is per-item.
- **Elicitation protocol.** See R2 — free-response and A/B forced-choice disagree on the same items.

Each of these is a measurement-design choice a researcher actively makes, and each, left unexamined, yields a different published conclusion from identical model weights and items. (Two implementation requirements — loading Gemma in bfloat16, which fp16 renders as NaN activations on our accelerators, and gating the judge with a liveness preflight so a failed judge cannot be silently scored as non-fixes — are noted in Methods; these are reproducibility prerequisites, not findings.)

R2. The effect is protocol-dependent (Qwen3-4B)

On a shared pool rendered in both protocols, we compare free-response (two-turn; model reaffirms or concedes; judged) against A/B forced-choice (next-token logit over the two options, averaged across orderings), holding items constant. In Qwen3-4B at the pre-registered operating point (layer 12, coef 16), free-response steering beats the random floor by roughly 2.5× (factual 32%, rimsky 28% vs random 12%), whereas under A/B **neither** real vector beats the floor (rimsky 24% ≈ random 22%; factual 6% < random). A pre-registered GEE (Binomial, exchangeable working correlation, clustered on item) gives a significant protocol × vector interaction (p = 0.0002). The directional hypothesis we pre-registered was not supported, but the random-control-grounded conclusion is robust: **steering efficacy is contingent on the elicitation protocol**, and the field-standard A/B/MCQ protocol is the one under which the effect vanishes.

R3. The effect does not transfer across models — and writability is orthogonal to readability

To rule out that non-Qwen-4B nulls were merely mis-tuned, we ran a per-model, **outcome-blind** operating-point sweep: the steering layer was chosen by contrast separation (Cohen's d, blind to fix-rate) and the coefficient by a coherence frontier (blind to fix-rate), then a single evaluation was run at the chosen point. Table 1 reports the six-model result.

Table 1. Readability (max Cohen's d across candidate layers) vs writability (free-response genuine-fix rate at the fair operating point; random-vector floor in parentheses). A/B is null in every model.

Model	Size	Readability *d*	Free-response fix (best real / random)	Writable?
Gemma-3-1B	1B	−0.01	7.3 / 9.5	no
Gemma-3-4B	4B	0.68	2.2 / 4.8	no
Gemma-3-12B	12B	1.24	4.6 / 3.5	no
Qwen3-1.7B	1.7B	0.30	9.2 / 13.3	no
Qwen3-4B	4B	**2.07**	**34.1 / 9.5**	**yes**
Qwen3-8B	8B	2.10	11.8 / 13.7	no

Two patterns emerge. **(i) Readability scales with size in both families** — Gemma-3: −0.01 → 0.68 → 1.24; Qwen3: 0.30 → 2.07 → 2.10 — so larger models encode the honest-vs-sycophantic distinction more linearly (a linear probe on Qwen3-8B activations separates the classes at d = 2.10). **(ii) Writability does not scale and is not explained by readability.** CAA beats the random floor in only one of six models (Qwen3-4B), the result is non-monotonic within a family (Qwen3-4B steerable, the larger Qwen3-8B not), and — the sharpest evidence — **Qwen3-4B (*d*=2.07, writable) and Qwen3-8B (*d*=2.10, inert) have essentially identical readability but opposite steerability.** The three highest-readability models include two (Qwen3-8B d=2.10, Gemma-3-12B d=1.24) that are completely inert. Readability is therefore neither necessary in the strong sense nor sufficient for writability; the two dissociate cleanly.

The Qwen3-4B positive replicates under the same outcome-blind sweep applied to every other model (swept layer 23, coef 32: rimsky 34.1% vs random 9.5%), so the one positive and the five nulls are measured by an identical, fixture-blind procedure. Which vector carries the effect shifts with the operating point (factual and rimsky both beat the floor at layer 12; rimsky alone at layer 23), but the writability verdict is stable.

R4. A/B forced-choice is null everywhere

Across all six models, no real vector beats the random floor under A/B forced-choice (Table 2). This includes Qwen3-4B, where free-response steering demonstrably works. Consequently, an evaluation conducted only in the A/B/MCQ protocol — the common default — would conclude that CAA does nothing for sycophancy in *any* of the six models, missing the single genuine free-response effect entirely. We note one honest limitation: the A/B baseline is partly degenerate for the Qwen models, which prefer the sycophantic option on ~100% of items at baseline and are never flipped, so the A/B null reflects an insensitive instrument as much as unchanged behavior — which is itself part of the argument against single-protocol evaluation.

Table 2. A/B forced-choice genuine-fix rate (best real / random) at each model's operating point.

Model	A/B fix (best real / random)
Gemma-3-1B	8.3 / 7.6
Gemma-3-4B	0.0 / 1.2
Gemma-3-12B	1.6 / 0.0
Qwen3-1.7B	0.0 / 0.0
Qwen3-4B	18.8 / 25.7
Qwen3-8B	0.0 / 0.2

Discussion

Readability is not writability, and scale widens the gap. The linear representation hypothesis holds that concepts appear as directions (Park et al. 2023); our readability axis measures exactly that, and it climbs sharply with model size in both families. Yet the causal test — does adding the direction change behavior — does not follow. Qwen3-4B (d=2.07) and Qwen3-8B (d=2.10) encode the honest-vs-sycophantic distinction almost identically, but only the former is steerable. This is a direct, cross-family instance of the point that "representations that are readable from activations may not be writable through them" (Buchan 2026): a linear correlate of a concept need not be a causal control knob, and scaling improves the correlate, not the control.

Single-protocol steering evaluation is unsafe. Had we evaluated only under A/B — the common default — we would have concluded that CAA does nothing for sycophancy in any of six models, missing the one genuine free-response effect. This is the steering-specific face of a general problem: models are poor forced-choice selectors (Zheng et al. 2023), and moving between multiple-choice and open-style formats changes conclusions (Myrzakhan et al. 2024; Molfese et al. 2025). Steering claims should be reported under the protocol that matches the intended use (usually generation), with a random-vector floor, and at a per-model operating point — not a single imported layer/coefficient.

What this does not claim. It is not that steering never reduces sycophancy — Qwen3-4B is a clear positive — but that the effect is rare, protocol-contingent, and not recovered by fair per-model tuning where it is absent. We characterize *where* and *whether*, not the circuit-level *why*; we localize neither the components that make Qwen3-4B writable nor those that leave Qwen3-8B inert. Identifying them — e.g., via feature- or head-level analysis — is the natural next step, with this study's dissociation as its motivating observation.

Limitations

- **Coefficient selection.** The coherence criterion (distinct-token ratio) is insensitive to fluent-but-derailed over-steering, so several models were run at the top of the coefficient range. Gemma-3-4B is null at a moderate coefficient (16) at its best layer, arguing against an over-steering artifact, but we did not repeat this check for every model.
- **Power.** Baseline-sycophantic *n* is low for Qwen3-8B under free-response (n=51), making that null the weakest-powered cell.
- **Operating-point asymmetry.** Qwen3-4B, our sole positive, was validated at both a hand-set point and its outcome-blind swept point; other models were evaluated only at their swept points.
- **A/B degeneracy.** The Qwen models prefer the sycophantic option on ~100% of A/B items at baseline and are never flipped, so the A/B null partly reflects an insensitive instrument rather than unchanged behavior — which is itself part of the argument, but limits strong behavioral claims from A/B alone.
- **Scope.** Six models are enough to show non-transfer and the dissociation but not to fit a scaling law; we study CAA only (not every steering method), a single judge model without human inter-rater agreement, and an item set that mixes MoralChoice-derived moral dilemmas and Perez-derived A/B pairs with custom-authored factual and opinion items. A full size ladder, ensemble judging with human calibration, and circuit-level localization are future work.

Conclusion

Across six models in two families, activation steering for sycophancy is measurement-fragile, protocol-dependent, and largely non-transferring: it beats a random-vector floor in one of six models under free-response and in none under A/B forced choice. The sycophancy direction becomes increasingly *readable* with scale while remaining, in most models, un-*writable* — the two axes are orthogonal, most starkly in a pair of models with near-identical readability and opposite steerability. Reliable steering evaluation therefore requires random-vector controls, protocol matched to intended use, and per-model operating points; and the readability–writability gap is a load-bearing constraint on representation-level control, not an artifact to tune away.

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